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MILITARY SPECIFICATION

LATHES, SHAFT TURNING, TURRET, VERTICAL BED AND SLANT BED, COMPUTER NUMERICALLY CONTROLLED (CNC)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers horizontal spindle, turret lathes of vertical and slant bed configurations. The covered lathes shall be used for shaft turning, with a computer numerical control unit for directing the machine functions from command data input by the media specified herein and from data contained in the control memory.

1.2 Classification. The lathe shall be of the following size. The size to be furnished shall be as specified (see 3.5 and 6.2).

	Size 1	Size 2	Size 3
Swing over cross slide	10	12	15
Basic center distance	26	36	40

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, 2163 Airways Boulevard, Memphis, Tennessee 38114-5051, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3416

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2. APPLICABLE DOCUMENTS.

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interferences.

MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

(Unless otherwise indicated, copies of Federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, (ATTN: Mr. Stephen Lowell), Philadelphia, PA 19111-5094.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

U. S. DEPARTMENT OF LABOR, OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

29 CFR 1910 - Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402-0001.)

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(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN GEAR MANUFACTURERS' ASSOCIATION (AGMA)

AGMA 390.03a - Gear Classification, Materials and Measuring Methods for Bevel, Hypoid, Fine Pitch Wormgearing and Racks Only as Unassembled Gears. (DOD adopted)

AGMA 2000-A88 - Gear Classification and Inspection Handbook. Tolerances and Measuring Methods for Unassembled Spur and Helical Gears. (DOD adopted)

AGMA 2001-B88 - Fundamental Rating Factors and Calculation Methods for Involute Spur and Helical Gear Teeth. (DoD adopted)

(Application for copies should be addressed to the American Gear Manufacturers' Association, Standards Department, 1500 King Street, Suite 201, Alexandria, VA 22314-2717.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B5.9 - Spindle Noses for Toolroom Lathes, Engine Lathes, Turret Lathes, and Automatic Lathes. (DOD adopted)

ANSI B11.6 - Safety Requirements for the Construction, Care, and Use of Lathes. (DOD adopted)

ANSI B93.114M - Pneumatic Fluid Power Systems- Standard for Industrial Equipment and Machine Tool Applications. (DOD adopted)

ANSI/ISO 8860/1 - Dimensional, Physical, and Magnetic Characteristics.

ANSI/ISO 8860/2 - Track Format.

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ANSI/ISO 9293 - Volume and File Structure of Flexible Disk Cartridges for Information Interchange.

ANSI/ISO 54 - Cylindrical Gears For General Engineering and for Heavy Engineering, Modules and Diametral Pitches of. (DOD adopted)

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018-3363.)

ELECTRONIC INDUSTRIES ASSOCIATION (EIA)

RS-227 - One-Inch Perforated Tape.

RS-232 - Data Terminal Equipment and Data Communication Equipment, Interface Between, Employing Serial Binary Data Interchange. (DOD adopted)

RS-244 - Character Control for Numerical Machine Tool Control Perforated Tape. (DOD adopted)

RS-267 - Axis and Motion Nomenclature for Numerically Controlled Machines. (DOD adopted)

RS-274 - Interchangeable Variable Block Data Format for Positioning, Contouring, and Contouring/Positioning Numerically Controlled Machines. (DOD adopted)

RS-358 - Subset of American National Standard Code for Information Interchange for Numerical Machine Control Perforated Tape. (DOD adopted)

RS-494 - 32 Bit Binary CL Exchange (BCL) Input Format for Numerically Controlled Machines.

(Application for copies should be addressed to the Electric Industries Association, 2001 I Street, N.W., Washington. DC 20006-1899.)

JOINT INDUSTRIAL COUNCIL (JIC)

JIC H-1 - Hydraulic Standards for Industrial Equipment and General Purpose Machine Tools.

(Application for copies should be addressed to the Joint Industrial Council, c/o The National Machine Tool Builders' Association, 7901 Westpark Drive, McLean, VA 22101-4269,)

NATIONAL ELECTRICAL MANUFACTURERS' ASSOCIATION (NEMA)

NEMA MG 1 - Motors and Generators. (DOD adopted)

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(Application for copies should be addressed to the National Electrical Manufacturers' Association, 2101 L Street, N.W., Washington, DC 20037-1580.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 79 - Electrical Standard for Industrial Machinery. (DOD adopted)

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269-9101.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First Article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.4) in accordance with 4.4.

3.2 Design. The lathe shall be new (not a prototype) and one of the manufacturer's current production models capable of operations in accordance with the requirements herein. The lathe shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be readily accessible for adjustment, replacement, or repair.

3.2.1 Reclaimed materials. The machine may contain reclaimed materials to the maximum extent possible provided such materials will not jeopardize the intended use, performance, or design life of the machine. Reclaimed materials shall have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used, rebuilt or remanufactured products are allowed under this specification.

3.2.2 Energy efficiency. The lathe and its components that directly consume energy in normal operation shall be designed and constructed for energy efficiency in accordance with the developments available within the industry.

3.2.3 Controls. All mechanical, electrical, hydraulic, and pneumatic operating controls shall be located convenient to the operator's work station(s).

3.2.4 Safety and health requirements. All parts, components, mechanisms, and assemblies furnished on the lathe, whether or not specifically required

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herein, shall conform to all requirements of OSHA 29 CFR 1910 and ANSI B11.6. If a conflict arises between 29 CFR 1910 and ANSI B11.6, 29 CFR 1910 shall apply. Additional safety and health requirements shall be as specified (see 6.2). Covers, guards, or other safety devices normally furnished as standard on the manufacturer's commercial machine supplied to the commercial market shall be provided for the point of operation and all other parts of the lathe that present safety hazards. The supplier of the machine shall provide the special guarding which shall meet OSHA 29 CFR 1910.

3.2.5 Environmental protection. During operation, service, transportation, or storage of the machine, it shall not emit materials hazardous to the ecological system as prohibited by applicable Federal, state, or local statutes in effect at the point of installation.

3.2.6 Lubrication. Means shall be provided to ensure lubrication for all moving parts. Recirculating lubrication systems shall include a cleanable or replaceable filter and a low lubrication level warning light that is in clear view of the operator's normal work station. An automatic lubrication system shall be provided for all ways and axis screws. Each lubricant reservoir shall have at least a 24-hour capacity and means for determining fluid level. All oil holes, grease fittings, and filler caps shall be readily accessible.

3.2.7 Interchangeability. All parts subject to replacement shall be manufactured to definite dimensions and tolerances. Replacement parts except for hand-fitted parts, such as ways, shall be interchangeable without requiring modification. Tooling and accessories normally supplied as standard shall also be considered replacement parts.

3.3 Construction. The lathe shall be constructed of parts which are new, without defects, and free of repairs. The structure shall withstand all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale, and mismatching. No processes such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Fastening devices. All fasteners shall be installed to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be permanently installed.

3.3.3 Surfaces. All surfaces shall be clean and free of harmful or extraneous materials. All edges shall be either rounded or beveled unless sharp- ness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.4 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only when specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

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3.3.5 Painting. Unless otherwise specified (see 6.2), the lathe shall be painted in accordance with the manufacturer's commercial practice and color.

3.3.6 Threads. All threaded parts used on the lathe and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Drive train gears. All gears of spindle and axis drive trains shall conform to or exceed all provisions of AGMA 2000-A88, AGMA 2001-B88 and AGMA 390.03a, for the inch-pound system of measurement or ANSI/ISO 54 for the metric system of measurement. The gears shall be manufactured of a suitable ferrous material. The gears shall be of proper width and size to transmit full-rated torque and horsepower throughout the speed ranges without failure for the expected service life of the machine. Gears in the drive train shall be hardened and ground steel. Working surface hardness shall be not less than Rockwell C-48.

3.3.8 Electromagnetic interference control. When specified (see 6.2), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystem(s) class and the emission and susceptibility requirements shall be as specified.

3.4 Components. The lathe shall consist of not less than a bed and base, numerically controlled headstock and tool slide mechanisms, a machine control unit and a closed loop feedback system. In addition, the machine shall be equipped with the type headstock spindle and tool turret required to provide the capabilities, tooling, and axes of motion required for the size machine specified under 1.2.

3.4.1 Bed and base. The bed configuration shall be of the slanted or vertical construction, having the longitudinal bedways running horizontally one above the other in a vertical plain, or on an angle upward and to the rear not more than 45 degrees from vertical to provide horizontal carriage movement along the longitudinal axis, and vertical or slanted cross slide movement. The bedway surfaces which support and guide machine slides shall have a surface hardness of not less than Rockwell C-50, or shall be anti-friction type. The base shall have a chip and coolant pan designed for holding a large volume of chips. When specified (see 6.2), an automatic chip removal system shall be provided. Means shall be provided for securing and leveling the machine on a foundation.

3.4.2 Headstock. The machine shall have a headstock that supports the spindle and spindle drive mechanism. The headstock and all components located therein shall be capable of utilizing the full machine horsepower without vibration or chatter. Unless otherwise specified (see 6.2) the spindle nose shall be of the size and type specified in table I. The spindle nose shall conform to ANSI B5.9 for the size and type provided, and shall rotate at the speeds specified in table I. All bearings shall be of a size, type, and precision class suitable for the application. Rotating components of the spindle drive system shall be lubricated by means of a splash system or a pressurized lubrication system.

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3.4.3 Turret. The machine shall be equipped with a disk or drum type turret, mounted on a cross slide and carriage unit in the vertical or slanted position. It shall have movement on the longitudinal axis and the cross axis. The carriage and cross slide movements shall operate as programmed for straight and contour turning on both internal and external work surfaces. Unless otherwise specified (see 6.2), the turret shall have at least six tool stations for mounting both internal and external cutting tools. Adjustable gibs shall be provided to compensate for wear between slide units. All turret functions, including modes of slide departure, feed selections and rapid travel, tool selections by turret indexing, and coolant on and off shall be numerically controlled from input media specified herein and by manually operated control devices. The axis and motion nomenclature shall be in accordance with EIA RS-267. The turret, when indexed, shall be accurately positioned and locked by a curvic coupling or a similar locking device having equal positioning and clamping capability. Slide movement shall be inhibited should a turret fail to seat fully when indexed. The turret shall be designed to permit use of pre-set or qualified tools or have means for accurately setting standard tooling. Tool offsets shall permit accurate tool point positioning. A built-in coolant system shall deliver coolant to the tool in the cutting position. All turret indexing and feed mechanism shall be protected from chips and dirt.

3.4.4 Feed mechanism. The drive unit and feedback device for each slide unit shall form a closed loop system capable of controlling slide movements in precise increments for accurate feed and positioning. Each carriage and cross slide shall be driven by an anti-backlash, precision recirculating ball screw and servo motor. Each ball screw shall be covered or shielded for protection from chips, dirt, and fluids. The response to command signals for modes of departure, feed speeds, rapid travel and start or stop shall be as required to produce workpart accuracies specified herein. Variable feed rate, variable spindle speed, or both shall provide a means for obtaining the optimum feed rate and spindle speed for part accuracy and surface finish. The feed back device shall be the manufacturer's standard equipment to accurately measure each slide movement and send signals to the control unit for correspondence of slide position and command data. Input resolution shall be not greater than 0.0001-inch or 0.002mm.

3.4.5 Tailstock. Unless otherwise specified, (see 6.2), the standard manually operated tailstock shall be supplied on the shaft turning machine to provide both chucking and shaft turning capability. The unit shall be designed for using interchangeable live centers or have a built-in anti-friction spindle. When specified (see 6.2), the spindle shall be power-operated by manual control and have adjustable spindle pressure. Means for moving the tailstock along its ways by power shall be provided. Way wipers or other means of preventing chips and damage causing materials from entering between the tailstock and ways shall be provided.

3.4.6 Computer numerical control (CNC). The lathe shall have a fully automatic, solid-state, integrated circuit type, computer numerical control. The machine control unit (MCU) shall provide automatic control of machine functions, operating modes, linear axes movement, spindle operation, turret

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rotation, and other part program-directed functions for lathes of the type specified herein. Control axes shall be identified in accordance with EIA Standard RS-267. The control shall have linear and circular interpolation and be capable of simultaneous control of two linear axes. Control features shall include an EIA RS-232 interface, buffer storage, fixed cycles, part program storage, part program and buffer edit capability, and control diagnostics. The MCU shall direct machine functions from part program data stored in memory. The data stored in memory shall be input by the media specified herein or manually input data (MDI) from a keyboard. The control shall initiate a halt or an error message should a fault condition occur in the control. All necessary executive program routines shall be furnished. A line voltage of ±10 percent from normal shall not adversely affect the numerical control system function. The control shall function in ambient temperatures ranging from 32 to 120 ^{o}F, and humidity ranging from 5 to 95 percent noncondensing. The control shall automatically shut down when internal operating temperature exceeds safe operating temperature. The control shall be inch/metric switchable with control resolution of 0.0001-inch/0.001mm for each axis.

3.4.6.1 Data input.

3.4.6.1.1 Punched tape data input. Unless otherwise specified (see 6.2), the input media for the control shall be a 1-inch, 8-channel punched tape. The data input format and codes shall comply with EIA Standards RS-227, RS-274, RS-358 and RS-244. The control shall be equipped with a photoelectric tape reader capable of holding, feeding, and reading tape prepared in the format specified at a rate not less than 150 characters per second. Unless otherwise specified, (see 6.2) the tape reader shall be supplied with standard feed and take-up spools 5-1/4 inches in diameter. The tape reader shall be housed in a dust free area of the numerical control cabinet and shall be easily accessible for cleaning and tape handling. The control shall have a tape error detection capability to continuously check for the following errors:

a. Odd parity per EIA RS-244.
b. Even parity per EIA RS-358.
c. Misaligned sprocket holes.
d. No sprocket holes.

Any of the above errors shall halt machine functions and illuminate an error light or a message on the data display screen to indicate a fault condition.

3.4.6.1.2 Binary cutter location. When 32 Bit Binary Cutter Location (BCL) Exchange Input for Numerically Controlled Machines is specified, (see 6.2) the input media shall be a 3 1/2-inch hardshell diskette structured in accordance with ANSI/ISO 8860/1, ANSI/ISO 8860/2, and ANSI/ISO 9293. The control shall recognize and act upon command data formatted in accordance with EIA RS-494. Section 10.1 shall apply to all BCL controls. Specific function set(s) required shall be provided as specified (see 6.2). Manufacturers providing functions or features that cannot be addressed by command codes currently covered by EIA RS-494 shall apply to the Electronic Industries Association, IE-31 Committee on Numerical Control Systems and Equipment for an extension of the standard codes to cover the new features or functions. The control shall

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be equipped with a 3-1/2 inch hardshell diskette reader. The reader shall have the capability to perform a cyclic redundancy check of the input data and be capable of reading not less than 15,000 characters per second.

3.4.6.1.3 Diskette input. When specified, (see 6.2) a portable diskette reader/writer shall be provided with all the hardware and software necessary to interface between the control and the reader/writer. The reader/writer shall receive and store part program data from the control unit and be capable of loading stored programs to the memory of the control unit. Part program storage shall be on a 3 1/2-inch hardshell diskette. The diskette shall meet the requirements of ANSI/ISO 8860/1, ANSI/ISO 8860/2, and ANSI/ISO 9293. The communication between the reader/writer shall be by means of EIA RS-232 communication ports.

3.4.6.1.4 Manual data input. A means for manual data input (MDI) shall be furnished to permit manual entry of alphabetical, numerical, and special characters necessary for control of all programmable machine functions. The manually input data shall be immediately displayed on the data display to permit data verification. The MDI shall be capable of entering a block of data into memory upon command.

3.4.6.2 Postprocessor. When specified (see 6.2), the system shall be furnished with a postprocessor program(s) to be utilized in off-line computer assisted program preparation. The postprocessor shall be provided for converting the specified program language (see 6.2) to the form and format required to operate the machine tool and control. The postprocessor shall be written for use on the type computer specified (see 6.2).

3.4.7 Standard control features.

3.4.7.1 Block delete. The block delete feature shall permit bypassing any blocks of information identified by a block delete code for use when a portion of a program is to be omitted.

3.4.7.2 Buffer storage. The MCU shall have buffer storage capable of storing one or more blocks of program data, read from memory or as input via MDI keyboard. As blocks are transferred to active command, new blocks shall be read into the buffer.

3.4.7.3 Display. Unless otherwise specified, (see 6.2) the display shall be a cathode ray tube (CRT). The CRT shall be capable of displaying machine system parameters and program codes from memory, buffer storage, and active data. The CRT shall be capable of displaying 256 characters or more.

3.4.7.4 Tool offsets. Control shall have not less than 32 pairs of program selectable tool offsets. The range of the offsets shall be not less than +99.9999 inches.

3.4.7.5 Program data edit. Part program edit capability shall be provided as means for editing tape and MDI input to optimize part program data. It shall be possible to modify part program data by adding, deleting, correcting, or completely changing stored data and operating with the edited program.

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3.4.7.6 Sequence number search. The operator shall be able to search through the part program data for a designated block number by initiating the search through the use of the MDI keyboard.

3.4.7.7 Jog. The jog control shall permit movement of the slide in the plus or minus direction of travel at preset jog rates such as high, low, or selectable rates of jog.

3.4.7.8 Diagnostic capability. The control shall include a diagnostic system for monitoring the control during operation and for locating malfunctions when troubleshooting. It shall bring the system to a controlled stop if a malfunction occurs during operation. The system shall give fault data in understandable code or message form on the display unit which will identify a problem at board level in the control and identify program errors.

3.4.7.9 Feedrate override. Means shall be provided to override the programmed feedrate from 0 to not less than 120 percent.

3.4.7.10 Programmed and optional stops. The control shall recognize programmed stopping points in the machining cycle. All controlled motions shall be stopped when the code for a programmed stop is read. The optional stop code shall be ignored unless the control is set to recognize the command.

3.4.7.11 Thread cutting. Single point thread cutting shall be provided for producing all standard straight and taper threads, multiple lead threads, and variable lead threads. Both internal and external threading shall be possible. The machine shall be capable of providing, by means of parts program, any thread pitch of 0.9999 inches and less, and any thread length of the capacity of the machine and less, unless otherwise specified (see 6.2).

3.4.7.12 Feed hold. The feed hold function shall permit the operator to interrupt feed rate and stop all motion. While in the feed hold condition, the machine may be operated manually. When normal operation is resumed, motions will resume at their previous feed rate.

3.4.7.13 Zero reference. Means shall be provided to establish a zero point at any position throughout the range of departure of each controlled axis. An operator control shall be provided to return the machine slides to the initial zero position for synchronizing the machine with the control.

3.4.7.14 Jog. Means shall be provided for jogging axis position manually. Movements in the plus and minus directions shall be selectable in increments of 0.0001, 0.001, 0.01, 0.1 and 1.0 inches on each axis. The jog capability shall function in each mode of operation except the automatic mode.

3.4.7.15 Constant surface speed. The constant surface cutting speed control shall permit maintaining a constant cutting speed within the limits of the machine, regardless of part diameter. Spindle speed control shall be based on the part diameter and the programmed surface speed to maintain a constant cutting rate.

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3.4.7.16 Feedrate programming. Feedrate shall be programmed in either inches per minutes (IPM) or inches per revolution (IPR). In the feed per revolution mode all axial feed movement shall be a function of the spindle rotation. Feed per minute shall be programmed directly in inches per minute.

3.4.7.17 Part program storage. Part program storage shall be either solid state, hard disk, or bubble type memory. The MCU shall be capable of storing in memory the equivalent of not less than 1000 feet of post processed part program tape.

3.4.7.18 Spindle speed override. A spindle speed override shall be provided to allow manual adjustment of spindle speed in either continuous adjustment or in steps of not greater than 10% of programmed speed. The spindle speed override function shall control not less than the range or override between 60% and 100% of programmed speed. The spindle speed override control shall have a position that allows return of speed to that which it was programmed.

3.4.8 Optional control unit features. The control unit shall be equipped with the optional features specified in 3.4.8.1 through 3.4.8.8. (See 3.9 for optional lathe attachments.)

3.4.8.1 Diameter programming. When specified (see 6.2), the lathe shall have diameter programming capable of accepting input information dimensioned in diameter rather than radii.

3.4.8.2 Decimal point programming. When specified (see 6.2), the lathe shall accept decimal point programming and shall recognize, and interpret numbers in a part program that contains a decimal point.

3.4.8.3 Dwell. If the control is required to provide a dwell period (see 6.2), the desired dwell time shall be programmed using a preparatory code. The programmable dwell time period shall be 1.0 second or less to 99.999 seconds or more.

3.4.8.4 Zero offset differential resolver. If zero offset by differential resolver is required to be furnished (see 6.2), the differential resolver shall be provided as additional means of establishing a reference point at any location on each axis of motion under numerical control.

3.4.8.5 Independent feedback. If independent feedback measuring system is required to be furnished (see 6.2), the feedback measuring system shall function as an independent unit so that accuracy is not affected by other components. It shall be of a type in which there is no physical contact between stationary and moving elements of the measuring device.

3.4.8.6 Computer graphics. When specified (see 6.2), computer graphic capabilities shall be provided as required and as fully described by the procuring activity.

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3.4.8.7 Conversational programming. When specified (see 6.2), conversational programming shall be provided as required and as fully described by the procuring activity.

3.4.8.8 Probing capabilities. When specified (see 6.2), the lathe shall have probing capabilities as required and as fully described by the procuring activity.

3.4.9 Coolant system. The coolant system shall include a sump or reservoir, a power driven pump, and all necessary piping. The sump or reservoir shall have sufficient capacity to permit full flow of coolant. The system shall have means for draining and cleaning and shall include a baffle or a strainer which is easily removable for cleaning. Means shall be provided to permit the operator to direct and control the amount of coolant over the entire work area.

3.4.10 Pneumatic system. The pneumatic system shall conform to the requirements of ANSI B93.114M.

3.4.11 Hydraulic system. The hydraulic system shall conform to or exceed the requirements of JIC H-1. The hydraulic system shall be complete, including all pumps, valves, piping, cylinders, and pressure controls. Means of protection from overpressure in high pressure lines shall be provided to prevent damage to components. A filter system shall be provided to ensure delivery of clean fluid. The hydraulic reservoir shall have means for determining fluid level, a provision for draining, and a cleanout access plate if the reservoir is not removable.

3.4.12 Electrical system. The electrical system shall be wired to conform to NFPA 79. Unless otherwise specified (see 6.2), each machine shall be wired to draw all of its electrical power from a single 230/460-volt, 3-phase, 60-Hertz (Hz) circuit. The lathe shall be initially wired for operation as specified (see 6.2), and have full instructions and circuit diagrams written in the English language as to how to convert voltage from one to another with all hardware included.

3.4.12.1 Motors. Motors shall meet the requirements of 3.4.12. Motors shall be rated for continuous duty and shall have ball or roller bearings of the sealed and permanently lubricated type. Unless otherwise specified (see 6.2), alternating-current (AC) motors shall be designed to operate on 60-Hz. The motor enclosure shall meet the requirements for a dripproof enclosure. All motors shall conform to the requirements of NEMA MG-1.

3.4.12.2 Worklight. Unless otherwise specified (see 6.2) a worklight shall be provided. The light shall have a protective shield, an on-off switch, and shall be adjustable to reflect maximum light on the work area. If a built-in lighting system is provided, the adjustable light described above shall not be required.

3.4.12.3 Hour meter. When specified (see 6.2), the lathe shall be equipped with an hour meter. The hour meter shall be installed to display accumulated operating time for the main drive motor. The meter shall be of the

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nonresetting type and shall have a range of 0 to 99,999 hours in increments of not greater than 1 hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the machine.

3.5 Size and capacity. Unless otherwise specified (see 6.2), the size and capacity of the lathe shall meet the requirements of table I for the machine specified. Machine capacity shall be not less than the stated requirements of the size ordered. When a range is given, the required performance shall be from the stated minimum or less to the stated maximum or greater.

3.5.1 Physical size limitations. When specified (see 6.2), the physical size of the machine shall not exceed the height, width, and length restrictions specified by the procuring activity to ensure the machine will fit in its future operating location.

3.6 Alignment tolerances. Each lathe shall meet the alignment tolerances specified in table III.

3.7 Performance. The lathe shall be capable of meeting the performance requirements of table II when tested using the procedures given in 4.8.5.

3.7.1 Additional performance requirements. Performance features required in addition to, or in lieu of the requirements herein shall be as specified and fully described (see 6.2).

3.8 Standard equipment. All standard equipment normally provided with the manufacturer's commercial machine shall be furnished.

3.9 Optional equipment. Optional equipment shall be furnished as specified and fully described (see 6.2).

3.10 Repair parts. Required repair parts shall be furnished as specified and fully described (see 6.2).

3.11 Marking on instruments, control panels, charts, and plates. All words on instruments, control panels, charts, and plates shall be in the English language. Characters shall be permanently marked in boldface on a contrasting background. All plates shall be corrosion resistant.

3.11.1 Lubrication plate. When specified (see 6.2), a lubrication plate shall be attached to each machine. Unless otherwise specified (see 6.2), the information provided on the plate shall be as listed below:

Points of lubricant application
Type of lubricant
Servicing interval
Military or Federal specification for each lubricant (if applicable)

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3.11.2 Nameplate. A nameplate shall be securely attached to each machine. Unless otherwise specified (see 6.2), the nameplate shall provide the information listed below:

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Amp rating of largest motor
Short circuit/over-current protection rating
Contract number or order number
National Stock Number or Plant Equipment Code
Date of manufacture

3.11.3 Hydraulic fluid requirements plate. When specified (see 6.2), a hydraulic fluid requirements plate shall be attached to the machine at the hydraulic reservoir fill point. Unless otherwise specified (see 6.2), the plate shall contain the following information:

Type of fluid (SAE or Military or Federal specification number)
Type of filter

3.12 Workmanship. Workmanship of the machine and its accessories shall meet all requirements specified herein and shall be of a quality equal to that prevailing among manufacturers producing equipment of the type covered by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submissions of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

c. Acceptance test (see 4.6).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed in accordance with the test conditions specified in 4.8.1.

4.4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.7 and all tests in 4.8. The machine shall pass the examination and all tests to be accepted.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2), quality conformance inspection shall consist of the examination in 4.7, the tests in 4.8.1 through 4.8.5 and the inspection in 4.9. The machine shall pass the examination, all tests, and the inspection to be accepted.

4.6 Acceptance test. Acceptance test shall be performed on each lathe to ensure conformance with this specification. Unless otherwise specified (see 6.2), the acceptance test shall be performed only after the machine is installed at its final location. The acceptance test shall consist of the examination in 4.7 and all tests in 4.8. The machine shall pass the examination and all tests to be accepted.

4.7 Examination. The machine shall be examined to determine compliance with all requirements of this specification.

4.8 Tests.

4.8.1 Test conditions. All tests shall be performed in an indoor facility with ambient conditions of 41 to 104 ^{0}F and 20 to 95 percent relative humidity.

4.8.2 Operational test. Each lathe shall be operated under numerical control for a period of four hours without load. The numerical control program shall cycle each axis through its entire range addressing all speeds, feeds, direction of rotation, and standard control functions specified in this document. Proper operation of all machine functions, controls, motors, adjusting mechanisms, and accessories shall be verified during the test period.

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4.8.3 Alignment tolerance test. Each machine shall be tested to determine compliance with the alignment tolerance requirements of 3.6. The alignment tolerance test shall be performed prior to and subsequent to the performance test given in 4.8.4 through 4.8.5.1 Any machine adjustments required to comply with 3.6 during or after the performance test shall be cause for rejection.

4.8.4 Performance accuracy tests. The machine shall be operated under CNC in each mode of operation to test the machine's response to command data. The test shall be conducted in accordance with the suppliers' standard test procedures to check control functions and machine response to commands for slide motions, turret indexing, spindle speed changes, feed rate changes and auxiliary functions. The safety limit stops shall be tested by attempting to exceed the limits of travel of the carriage and cross slide. All disks, instruments, materials, and tools required to perform and evaluate the tests shall be furnished by the supplier. All performance tests shall be completed without making any adjustments to or performing any maintenance on the machine.

4.8.5 Performance test. Two identical parts shall be machined to test accuracies and surface finish capabilities. The material shall be SAE 1045 or equivalent. Tooling design and geometry shall be of the manufacturer's choice provided carbide inserts and tool holders are of a negative rake configuration. Spindle speeds, depth of cut, and feed rates shall be appropriate to achieve the cubic stock removal rate specified in table II for the size machine required. The test shall include straight turning, taper turning, shoulder and face cuts, arcs, and single point threading. Surface finish on all cuts shall be 60 microinches or better. The test pieces shall meet the accuracies of table II.

4.8.5.1 Additional performance tests. Performance tests required in addition to or in lieu of the requirements given in 4.8.4 and 4.8.5 shall be as specified and fully described (see 6.2).

4.8.6 Electromagnetic interference control test. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.3.8 using the procedures given in MIL-STD-462.

4.8.7 Environmental protection certification of conformance. The contractor shall certify and maintain substantiating evidence that the product offered complies with all environmental protection restrictions in 3.2.6. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

4.9 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Packaging requirements. Unless otherwise specified, preservation, packing, and marking shall be in accordance with ASTM D 3951. When specified

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(see 6.2), level A preservation, level A or level B packing, and marking shall be accomplished in accordance with MIL-M-18058.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The machines covered by this specification are for manufacturing a variety of parts where numerically controlled lathes are advantageous. The machine will automatically machine parts held in a chuck or between centers.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Size required (see 1.2).

c. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

d. First article, if required (see 3.1).

e. Additional safety and health requirements, if required (see 3.2.4).

f. Painting, if different (see 3.3.5).

g. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.8).

h. Automatic chip removal system, if required (see 3.4.1).

i. Spindle nose, if different; specify optional spindle nose, if required (see 3.4.2).

j. Quantity of tools in tool turret, if different (see 3.4.3).

k. Tailstock, if not required (see 3.4.5).

l. Power operated tailstock, if required (see 3.4.5).

m. Input media, if other than 1-inch perforated tape (see 3.4.6.1.1.).

n. Take up reels, if different (see 3.4.6.1.1).

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o. Specify Binary Cutter Location (BCL), if required (see 3.4.6.1.2)

p. If BCL is required, specify the function sets of EIA RS-494 that are applicable to the machine being procured (see 3.4.6.1.2). (NOTE: SHOULD ASSISTANCE BE REQUIRED IN COMPLETING FUNCTION SET REQUIREMENTS, CONTACT DIPEC-SSM)

q. Specify diskette input, if required (see 3.4.6.1.3).

r. Specify programming language and type of computer, if postprocessor is required (see 3.4.6.2).

s. Display, if different (see 3.4.7.3).

t. Thread cutting capacity, if different (see 3.4.7.11).

u. Diameter programming, if required (see 3.4.8.1).

v. Decimal point programming, if required (see 3.4.8.2).

w. Dwell, if required (see 3.4.8.3).

x. Zero offset differential resolver, if required (see 3.4.8.4).

y. Independent feedback, if required (see 3.4.8.5).

z. Computer graphics, if required (see 3.4.8.6).

aa. Conversational programming, if required (see 3.4.8.7).

bb. Probing capabilities, if required (see 3.4.8.8).

cc. Electrical system, if different (see 3.4.12).

dd. Specify initial voltage (see 3.4.12).

ee. Motors, if different (see 3.4.12.1).

ff. Worklight, if not required (see 3.4.12.2).

gg. Hour meter, if required (see 3.4.12.3).

hh. Size and capacity, if different (see 3.5).

ii. Physical size limitations, if required (see 3.5.1).

jj. Additional performance features, if required (see 3.7.1).

kk. Specify and fully describe optional equipment, if required (see 3.9).

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ll. Specify applicable repair parts, if required (see 3.10).

mm. Lubrication plate, if required (see 3.11.1).

nn. Information to be provided on lubrication plate, if different (see 3.11.1).

oo. Nameplate, if different (see 3.11.2).

pp. Hydraulic plate, if required (see 3.11.3).

qq. Hydraulic plate, if different (see 3.11.3).

rr. Inspection conditions, if different (see 4.3).

ss. First article inspection, if different (see 4.4).

tt. Quality conformance inspection, if different (see 4.5).

uu. Acceptance test, if different (see 4.6).

vv. Additional performance test, if required (see 4.8.5.1).

ww. If preservation, packing, and marking in accordance with MIL-M-18058 is required, specify level required and specification number (see 5.1).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/ provided and that the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
6.3.1	DI-TMSS-80527	Commercial Off-the-Shelf (COTS) Manuals	Require all data be in the English language.
6.3.1	DI-TMSS-80528	Supplemental data for Commercial Off-the-Shelf Manuals.	Require all data be in the English language.

The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data

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Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.3.1 Technical data. All technical data ordered using a DD Form 1423 should be written in the English language.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a first article item or a standardproduction item from the contractor's current inventory and the number of items to be tested as specified in 4.4. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Definitions.

6.5.1 Inch-pound units. Inch-pound units are a system of measures based on the yard and pound commonly used in the United States of America and defined by the National Institute of Standards and Technology. Inch-pound units having the same names in other countries may differ in magnitude.

6.5.2 Metric units. Metric units are a system of basic measures defined by the International System of Units based on "Le Systeme International D'Unites (SI)," of the International Bureau of Weights and Measures. These units are described in ASTM E 380 and IEEE 268.

6.6 Measurement system. In this specification, all measurements, dimensions, sizes, and capacities are given in inch-pound units. These measurements may be converted to metric units through the use of the conversion factors and methods specified in FED-STD-376.

6.7 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.8 Cross-reference of classification changes. Classification changes from the previous issue of this specification are as follows:

MIL-L-80219A	MIL-L-80219B
Type I	Delete
Type II	Delete
Type III	Delete
Style A	Delete
Style B	Delete
Style C	Delete
Style D	Delete

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Size 1	Add
Size 2	Add
Size 3	Add

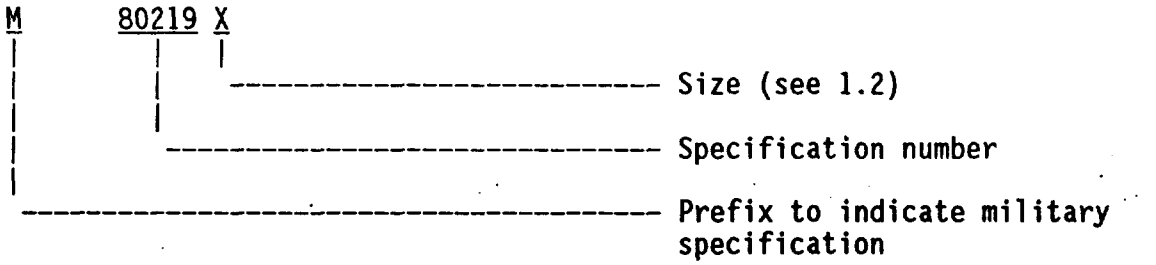
6.9 Safety and health requirements. 29 CFR 1910 limits only the total hazard level (noise, radiation, electromagnetic emissions, noxious vapors, air contaminants, and heat) of the environment in which a machine will operate. It does not limit the hazard level of individual machines in an operating environment. The procuring activity should analyze the existing hazard level in the proposed operating environment and specify additional requirements necessary to integrate this new machine into its future environment.

6.10 Training. Training required by the procuring activity should be provided as specified in the contract.

6.11 Warranty. Warranty requirements should be as specified by the procuring activity in the contract.

6.12 Inspection location. The contractor should identify in his response to the solicitation the location where inspection and tests are to be performed.

6.13 Part or Identifying Number (PIN). The PIN to be used for machines acquired to this specification are created as follows:



6.14 Subject term (keyword) listing.

Centers
Horizontal Lathe
Shaft
Turning

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TABLE I. Sizes and capacities.

Characteristics	Requirements		
	Size 1	Size 2	Size 3
Chuck size, inches	10	12	15
Swing over cross slide, inches	10	12	15
Distance between centers, inches	26	36	40
Spindle nose size and type	6"-A1	8"-A1	11"-A1
Horsepower, main drive motor	15 hp	30 hp	40 hp
Spindle speed range, RPM	120-1900	54-1500	33-850
Feed range, programmable feeds, IPM	0.1-150	0.1-150	0.1-125
Travel distances:			
X-axis, inches	7	5	10
Z-axis, inches	26	36	40
Optional features:			
Chuck size, inches	12	15	18, 21
Spindle nose	6"-A2	8"-A2	11"-A2, 15"-A1
Spindle speed range, RPM	120-2000	40-2000	20-1000, 30-1400
Horsepower, main motor	20 hp	40 hp	50 hp

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TABLE II. Performance accuracy.

Test to be performed	Requirement
Straight turning test, outside diameters:	
Turned diameter cylindrical within	0.0005 in 6"
Turned diameter round within	0.0003 TIR
Facing cut, flatness, concave only	0.0005 in 6" diameter
Taper turning	0.001 in./12 in.
Thread lead error	0.0005 in./12 in.
Shoulder cuts and facing cuts square with workpiece axis	0.0005 in./12 in.
Circular path, deviation from program path	±0.001 in./12 in.
Surface roughness, height not to exceed	60 microinches
Cubic stock removal rate:	
size 1	14 cu. in./min
size 2	30 cu. in./min
size 3	40 cu. in./min

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TABLE III. Alignment accuracies.

Test	Permissible error
Headstock spindle center runout	0.0004 in. TIR
Spindle nose runout	0.0003 in. TIR
Spindle taper runout:	
at end of spindle nose	0.0003 in. TIR
at end of 12-inch test arbor	0.0006 in. TIR
Spindle cam action	0.0003 in. TIR
Headstock alignment with bedways:	
Vertical plane	0 to 0.0005 in./12 in.
Horizontal plane	0 to 0.0005 in./12 in.
X axis square to Z axis	0.0015 in. /6 in.
Z axis parallel to spindle axis	0.0002 in. /12 in.
Turret indexing accuracy	±15 seconds of arc
Turret repeatability	±3 seconds of arc
Carriage positioning to command data	±0.001 in. /48 in. and 0.0002 in each additional 12 in.
Carriage positioning repeatability	±0.0005 in.
Cross slide positioning to command data	±0.00032 in.
Cross slide positioning repeatability	±0.0001 in.
Headstock and tailstock center alignment, vertical	0 to 0.001 in.
Tailstock spindle alignment:	
vertical, high at end of spindle, when fully extended	0.001 TIR
horizontal, forward at end of spindle, when fully extended	0.0005 in. TIR
Tailstock way alignment	0.0005 in./48 in.
Tailstock taper alignment:	
horizontal, at end of 12-inch test bar	0 to ±0.0005 in. TIR
vertical, at end of 12-inch test bar	0 to ±0.0005 in. TIR

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Custodians:
Army - AL
Navy - SH
Air Force - 99

Review activities:
Air Force - 84
DLA-GS

User activities:
Army - AR
Navy - YD, MC

Preparing activity:
DLA-IP

Project (3416-0232)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-L-80219B	901031

3. DOCUMENT TITLE
LATHES, SHAFT TURNING, TURRET, VERTICAL BED AND SLANT BED, COMPUTER NUMERICALLY CONTROLLED

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
Defense Industrial Plant Equipment Center ATTN: DIPEC-SSM	(1) Commercial 901-775-4791	(2) AUTOVON AV 683-4791
c. ADDRESS (Include Zip Code) 2163 Airways Blvd. Memphis, TN 38114-5051	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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